

**Sri Lanka Institute of Information
Technology**



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Web-Based Supermarket Ordering System

Final Report

Name	Student ID Number
Jayalath J.G.H	IT24101355
Alahakoon B.M.B.S	IT24101474
Dinod P.A.S	IT24101334
Sanvidu M.G.M	IT24101458
Fernando S.A.W	IT24101304
Thahani N	IT24104344

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1.Introduction

1.1 Project Overview

QuickMart is a web-based supermarket ordering system designed to modernize the shopping experience by allowing customers to browse products, place orders, make secure payments, and track deliveries online. It also enables staff to manage inventory, promotions, and delivery operations efficiently through a centralized platform.

The system aims to **digitize supermarket operations, enhance customer convenience, and improve staff productivity** while ensuring secure, fast, and reliable service for all users.

1.2 Objectives

The primary objectives of the system are to:

1. **Enable Online Shopping:** Allow customers to browse and purchase products conveniently through a web-based platform.
2. **Streamline Order Management:** Simplify order placement, tracking, and fulfillment for both customers and staff.
3. **Ensure Secure Payments:** Integrate reliable payment gateways to support safe and transparent online transactions.
4. **Optimize Inventory Control:** Provide real-time stock updates and low-stock alerts to maintain product availability.
5. **Improve Delivery Efficiency:** Manage delivery scheduling, route optimization, and status updates effectively.
6. **Support Multi-User Roles:** Assign specific access levels for admins, managers, staff, and customers for better control.
7. **Automate Promotions and Discounts:** Enable marketing managers to create and manage special offers and campaigns.
8. **Generate Analytical Reports:** Provide insights on sales, payments, and customer activity for data-driven decisions.
9. **Promote Operational Efficiency:** Reduce manual workload, improve coordination, and support the supermarket's digital transformation.

1.3 Target Users and Stakeholders

The system is designed for the following key stakeholders:

1. Customers

- Browse and purchase products online.
- Make secure payments and track orders.
- Provide feedback and reviews to improve services.

2. Admin Officer

- Manages user accounts, access permissions, and system reports.
- Monitors overall system activity to maintain integrity and security.

3. Inventory Manager

- Updates product information, stock levels, and prices.
- Handles low-stock alerts and supplier restocking operations.

4. Marketing Manager

- Creates and manages promotional campaigns
- Analyzes customer trends to boost engagement and sales.

5. Payment Manager

- Oversees all payment transactions and ensures secure processing.
- Handles payment confirmations, refunds, and issue resolution.

6. Delivery Coordinator

- Assigns delivery staff, manages routes, and tracks delivery progress.
- Updates order status and ensures timely delivery to customers.

1.4 Scope and Limitations

Scope:

1. **Online Shopping:** Customers can browse, search, and purchase products through an easy-to-use web interface.
2. **Secure Payments:** Integration with reliable payment gateways to process online transactions safely.

3. **Order and Delivery Management:** Real-time order tracking, delivery scheduling, and notifications for customers.
4. **Inventory Management:** Staff can add, update, or remove products and monitor stock levels accurately.
5. **Promotions and Discounts:** Marketing managers can create and manage promotional campaigns and special offers.
6. **User Management:** Role-based access for admins, staff, and customers to ensure data integrity and control.
7. **Reporting and Analytics:** Generation of sales, inventory, and performance reports for management insights.
8. **Multi-Device Accessibility:** The system can be accessed from desktops, laptops

Limitations:

1. **Third-Party Integration Risks:** Payment gateway reliability depends on external service providers.
2. **Device Compatibility:** Older browsers or devices may not fully support all system features.
3. **Language Support:** The initial version may support only English
4. **System Maintenance:** Requires regular technical updates and monitoring to prevent downtime.
5. **Data Security Threats:** Despite encryption and authentication, potential cyber risks remain.

2. Requirements

2.1 Functional Requirements

1. User Management

- Users can register, log in, and manage their accounts securely.
- Admins can add, update, or delete user accounts.
- Role-based access control must be implemented for different user types (Admin, Delivery Coordinator, Customer, etc.).
- The system should allow account recovery.

2. Product and Inventory Management

- Inventory managers can add, edit, or remove product details (name, price, description, category, stock quantity).
- The system should display real-time product availability to customers.
- Low-stock alerts must be automatically generated for inventory staff.
- Bulk product import/export should be supported (via Excel/CSV).

3. Shopping and Order Management

- Customers can browse or search for products by category, name, or price.
- Customers can add items to their shopping cart and update or remove them.
- The system should calculate the total cost, including taxes and discounts.
- Customers can confirm and place orders online.
- Order history and tracking details must be available to users.

4. Payment Processing

- The system must integrate with secure payment gateways (e.g., PayPal).
- Support for multiple payment methods (credit/debit cards, COD, etc.).
- Generate receipts for successful payments.
- Admins or payment managers can view and manage transaction logs.

5. Delivery Management

- Delivery coordinators can assign delivery staff and update delivery statuses.
- Customers should be able to track their order delivery progress.

6. Promotions and Marketing

- Marketing managers can create, edit, and remove promotional campaigns
- The system should apply valid promotions automatically during checkout.
- Customers should receive promotional emails and notifications.

2.2 Non-Functional Requirements

1. Performance

- The system should load all main pages (home, product listing, cart) within **3–5 seconds** under normal load.
- It must support **concurrent access** by multiple users without performance degradation.
- Database queries should be executed efficiently to handle real-time stock updates and transactions.

2. Security

- All sensitive data (passwords, payments, personal info) must be **encrypted** in storage and during transmission.
- Implement **secure login authentication** and **role-based access control**.
- Integrate **HTTPS** to ensure secure communication between client and server.
- Implement **automatic logout** after a period of inactivity.
- Maintain **audit logs** for all administrative actions.

3. Usability

- The user interface should be **intuitive, responsive, and user-friendly**.
- Navigation menus, buttons, and icons must be clearly labeled and easy to access.
- The system should support **multi-device compatibility** (desktop, laptop, tablet, smartphone).
- Include help sections, tooltips, or user guidance for first-time users.

4. Availability

- The system should maintain at least **99% uptime** during operational hours.
- Implement regular **data backups** and **recovery mechanisms** to prevent data loss.
- The application must be accessible 24/7, except during scheduled maintenance.

5. Maintainability

- The system architecture must support **modular design** for easier updates and bug fixes.
- Code should be **well-documented** and follow standard naming conventions.
- Error logs must be maintained to identify and resolve issues efficiently.

6. Scalability

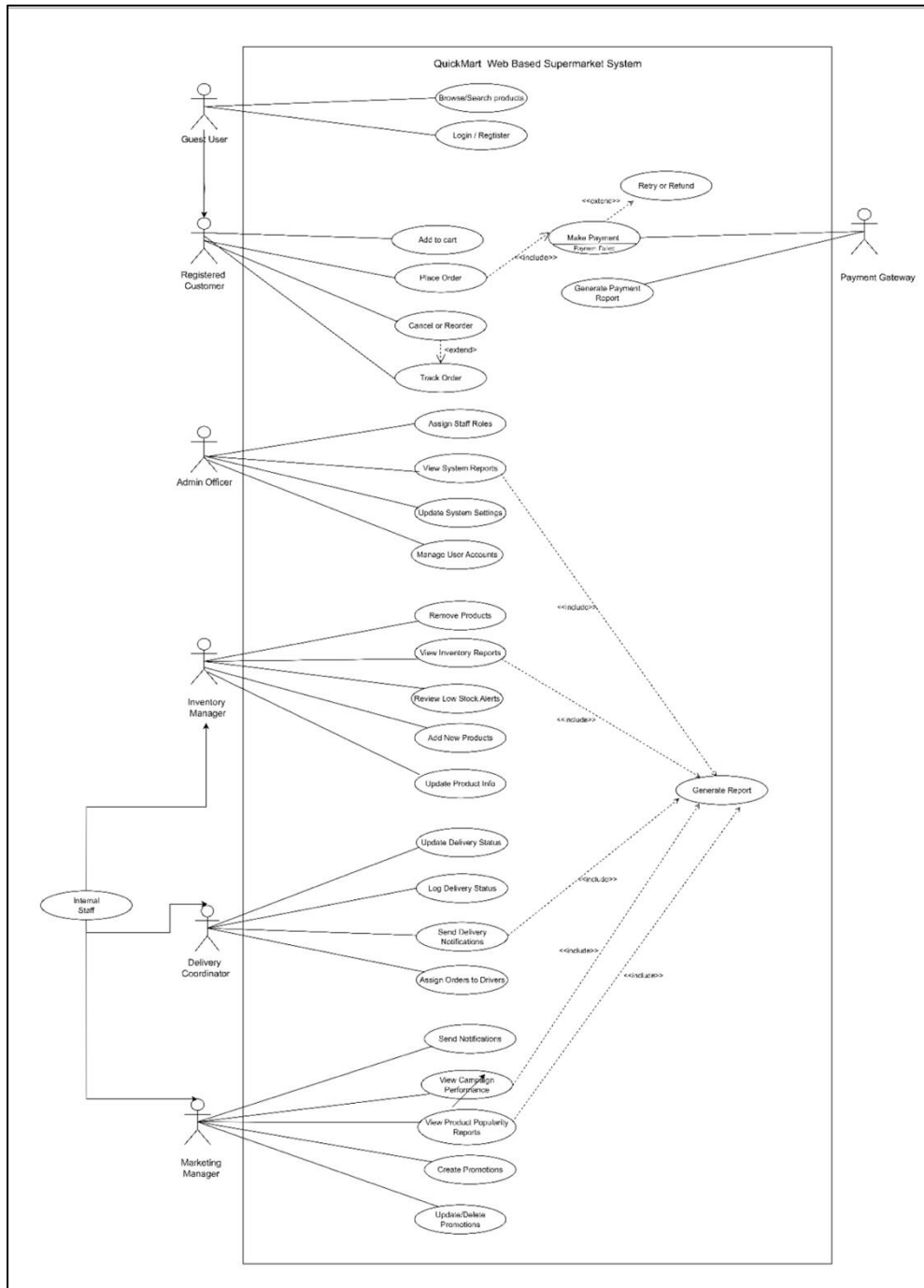
- The system should be scalable to support future growth in **users, products, and transactions**.
- It should allow integration with new payment methods or delivery systems without major redesigns.

2.3 Constraints and Assumptions

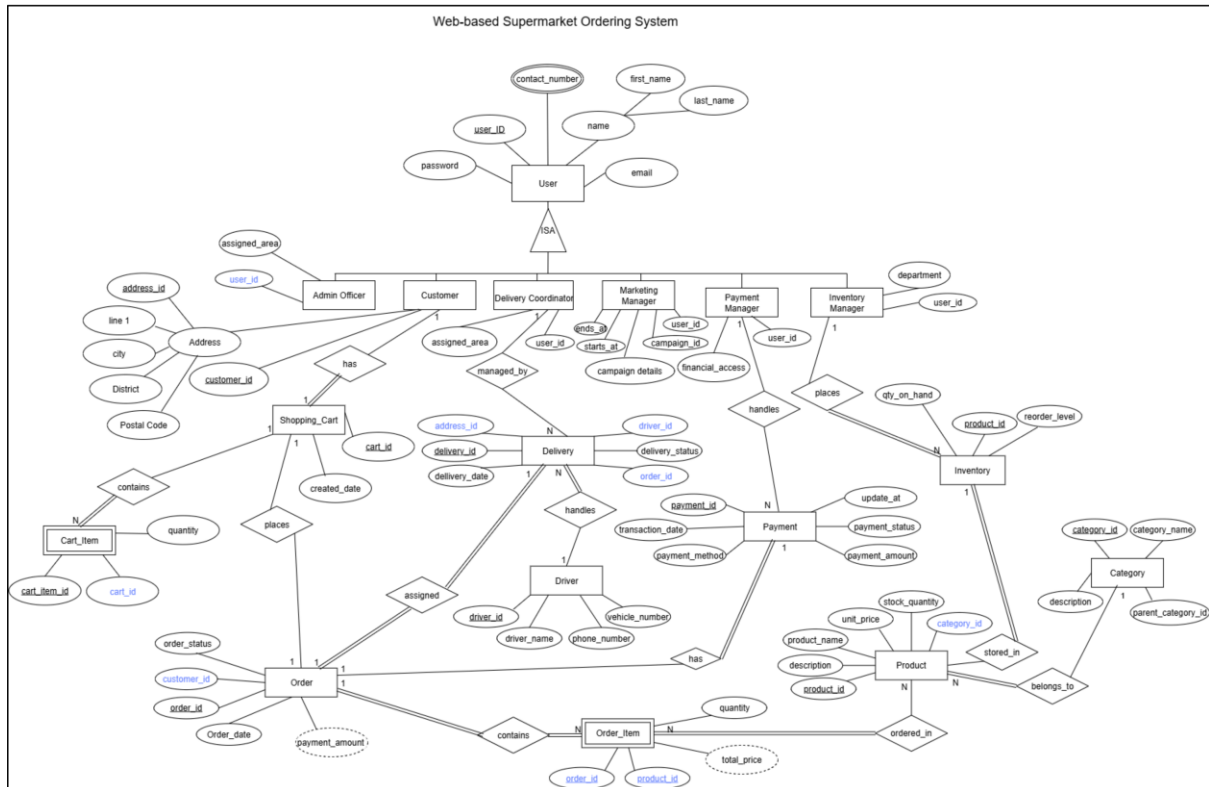
- **Platform Constraint:** The system is a web-based application.
- **Device Type:** Accessible via laptops, desktops
- **Assumptions:** Users have basic web literacy; users have a valid email/ for sign-up
- **Performance Limits:** Designed for small to medium user loads; may slow during high concurrent usage.
- **Security Limits:** Relies on users creating strong passwords.

3. Design

3.1 Use Case Diagram

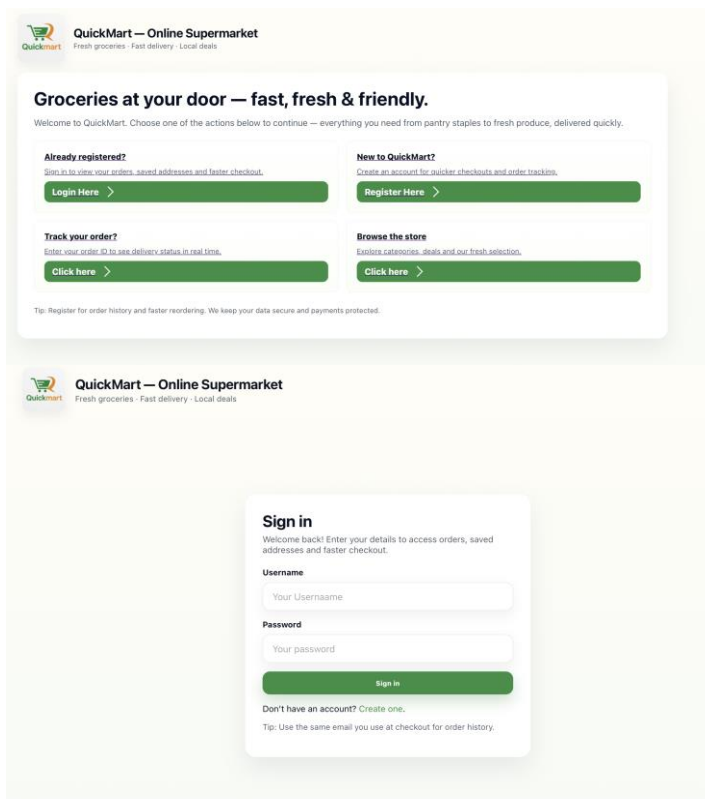


3.2 Database Design / ER Diagram



3.3 UI Sketches/Screenshots and Descriptions

Welcome Page



Administrator

QuickMart

Export CSVLogout

User: gayo

Admin Dashboard

Showing 8 user(s)

Add / Update User

Username

Email

Password (leave blank to keep)

Customer

Save

User List

Q Search username or email

RoleAll Roles

Clear

ID	USERNAME	EMAIL	ROLE	ACTIONS
1	gayo	admin@123	Administrator	Edit Delete
2	sach	sach@test	Inventory_Manager	Edit Delete
3	mani	mani@test	Payment_Manager	Edit Delete
4	shaa	shaa@test	Customer	Edit Delete
5	sisi	sisi@test	Delivery_Coordinator	Edit Delete
6	thani	thani@test	Marketing_Manager	Edit Delete
13	Chamodh	test@123	Customer	Edit Delete
14	test	test@123	Customer	Edit Delete

Inventory Manager

QuickMart · Inventory Manager

Download ReportLogout

User: sach

7

Visible SKUs

0

Low Stock (≤ 5)

0

Out of Stock

Add / Update Product

Product Name

Category

0.0

0

Image URL

Save Product

Products

Q Search products or categories

CategoryAll

StockAll

ID	IMAGE	NAME	CATEGORY	PRICE	QUANTITY	ADDED BY	CREATED	ACTIONS
1		Chocolate Cream Biscuits 400g	Biscuits	337.0	96	sach	2025-10-15 14:40	Edit Delete
2		Ginger Beer Pet Bottle 1.5L	Drinks	300.0	95	sach	2025-10-15 14:41	Edit Delete
3		Full Cream Fresh Milk Tetra 1L	Diary	500.0	97	sach	2025-10-15 14:41	Edit Delete
4		Watawala Tea Kahata Pouch 400g	Tea	855.0	99	sach	2025-10-15 14:42	Edit Delete

Payment Manager

QuickMart · Payment Manager

Generate Report

Logout

TOTAL ORDERS

3

TOTAL SALES

Rs 1,000

PENDING

1

REFUNDS

1

Search orders by id, customer, or amount...

All Statuses

All Types

Orders Overview

ORDER ID	CUSTOMER	AMOUNT	PAYMENT TYPE	PAYMENT STATUS	DELIVERY STATUS	ACTIONS	
1	shaa	1000.0	PAYPAL	APPROVED	DELIVERED	Approve	Refund
2	sisi	2374.0	CARD	REFUNDED	UNASSIGNED	Approve	Refund
3	shaa	2674.0	CARD	PENDING	PENDING	Approve	Refund

Saved Payment Methods

PAYMENT ID	USER	TYPE	CARD HOLDER	CARD NUMBER	EXPIRY	CUSTOM MESSAGE	ACTIONS
No payment methods found.							

Delivery Coordinator

QuickMart

Generate Report

Logout

Orders Overview

Assign drivers and update delivery statuses.

Status:

PENDING

ASSIGNED

SHIPPED

DELIVERED

Status: All

Driver: All Drivers

ID	CUSTOMER	TOTAL	PAYMENT	PAYMENT STATUS	DELIVERY STATUS	ADDRESS	DRIVER	ACTIONS
1	shaa	1000.0	PAYPAL	APPROVED	UNASSIGNED	15 Colombo - 00321	-- Select Driver --	Assign
2	sisi	2374.0	CARD	REFUNDED	UNASSIGNED	15 Mill Road - 00321	-- Select Driver --	Assign
3	shaa	2674.0	CARD	PENDING	PENDING	15 Colombo - 00321	-- Select Driver --	Assign

Driver Overview

Add new drivers and update their status.

Driver Name Contact Number Vehicle Number Add Driver

ID	NAME	CONTACT	VEHICLE	STATUS	UPDATE	
3	driver1	0702958518	WN-2295	AVAILABLE	AVAILABLE	Update Remove

Customer

QuickMart

Search fresh, fast.

1

1

Groceries, perfected.

Less friction. More flavor. Delivered with intention.

Bringing you the best products from around the world at the best prices

Shop now



Saver 4

Await for Promotional materials

All

On Sale

Biscuits

Drinks

Diary

Tea

Water

Seeds

ALL PRODUCTS

Reset



Rs 337.0

Add



Rs 300.0

Add



Rs 500.0

Add



Rs 855.0

Add

Drinking Water Pet 1.5L

Rs 130.0

Add

Marketing Manager

QuickMart Marketing

+ New Promotion

Export CSV

Promotions

Create, prioritize, and toggle the homepage carousel.

2

Total

1

Active

1

Inactive

Search title or description

Active

All

Sort

Priority (0 first)

ID	TITLE	DESCRIPTION	IMAGE URL	ACTIVE	PRIORITY	ACTIONS
1	Title Title Under Construction	Description... Await for Promotional materials	/images/promo/pic.jpg —	Yes	0	Edit Delete
No promotions found.						

New Promotion

Title

e.g., Weekend Saver

Description

Short, punchy copy for the carousel

Image URL (optional)

/images/promo/weekend.jpg or https://...

Priority (1 = highest)

1

Active?

☐

Create

Cancel

12

4.Implementation

4.1 Tools and Technologies Used

- Frontend: HTML
- Backend: Java
- Database: mysql workbench

4.2 Key Features Developed

1. **User Registration and Authentication** – Secure login and role-based access for admins, staff, and customers.
2. **Product Browsing and Search** – Customers can search and filter products by name, category, or price.
3. **Shopping Cart Management** – Add, update, or remove items from the cart with automatic total price calculation.
4. **Order Placement and Tracking** – Customers can place orders and track their delivery status in real time.
5. **Online Payment Integration** – Secure transactions through integrated payment gateways with confirmation receipts.
6. **Inventory Management** – Inventory Manager can add, edit, or remove products and monitor stock levels with low-stock alerts.
7. **Delivery Management** – Delivery coordinators can assign delivery staff, manage routes, and update delivery status.
8. **Promotions and Discounts** – Marketing managers can create and manage discount codes, bundles, and seasonal offers.
9. **Report Generation** – Admins can generate detailed sales, payment, and performance reports in CSV format.

5.Project Management

5.1 Agile Approach and Sprint Summary

Sprint planning

Sprint 1 (Week 1 - Week 3)

Sprint Completion Rate: ~100%

Sprint ID	User Story	Priority
SP1-01	PBI-01 – As a Customer, I want to register and log in so that I can access my account securely.	High
SP1-02	PBI-02 – As a Customer, I want to browse available products so that I can view items before ordering.	High
SP1-03	PBI-03 – As an Admin Officer, I want to manage user roles so that I can control access to admin functions.	High

SP1-04	PBI-04 – As an Admin Officer, I want to update product categories so that the store layout remains organized.	Medium
SP1-05	PBI-05 – As a Developer, I want to set up the project structure and database connections so that all modules can integrate smoothly.	High

Sprint 2 (Week 4 - Week 6)

Sprint Completion Rate: ~100%

Sprint ID	User Story	Priority
SP2-01	PBI-06 – As a Customer, I want to add products to my cart so that I can place orders easily.	High
SP2-02	PBI-07 – As a Customer, I want to edit or remove items from my cart so that I can manage my purchases.	High
SP2-03	PBI-08 – As a Marketing Manager, I want to create and manage promotional offers so that I can attract more customers.	Medium
SP2-04	PBI-09 – As an Inventory Manager, I want to add and update stock details so that product availability stays accurate.	High
SP2-05	PBI-10 – As an Admin Officer, I want to view all registered users so that I can manage their accounts effectively.	Medium

Sprint (Week 7 - Week 9)

Sprint Completion Rate: ~100%

Sprint ID	User Story	Priority
SP3-01	PBI-11 – As a Customer, I want to make secure online payments so that I can complete my purchase quickly.	High
SP3-02	PBI-12 – As a Customer, I want to view my previous orders so that I can track my purchase history.	Medium
SP3-03	PBI-13 – As a Marketing Manager, I want to schedule and display discount banners so that promotions appear at the right time.	Medium
SP3-04	PBI-14 – As an Inventory Manager, I want to receive alerts for low-stock products so that I can restock on time.	High
SP3-05	PBI-15 – As a System Admin, I want to monitor user and transaction logs so that I can ensure data integrity.	Medium

Sprint 4 (Week 10 - Week 12)

Sprint Completion Rate: ~100%

Sprint ID	User Story	Priority
SP4-01	PBI-16 – As a Customer, I want to receive email or SMS notifications for order confirmations and deliveries.	High
SP4-02	PBI-17 – As an Admin Officer, I want to generate sales and customer reports so that I can analyze performance.	Medium
SP4-03	PBI-18 – As a Developer, I want to integrate all modules and ensure smooth backend–frontend communication.	High
SP4-04	PBI-19 – As a Customer, I want to provide feedback and ratings so that I can share my shopping experience.	Medium
SP4-05	PBI-20 – As a QA Tester, I want to perform full system testing and fix bugs before deployment.	High

5.2 Task Distribution Among Team Members

Member Name	Scrum Role	Main Responsibility
J.G.H.Jayalath	Product Owner	Implement user authentication, and account management
Fernando S.A.W	Scrum Master	Develop online ordering and shopping cart features
Alahakoon B.M.B.S	Developer	Monitor stock levels and ensure accurate product availability.
Thahani N	Developer	Create and manage discount codes, bundle offers, and seasonal campaigns.
Sanvidu M.G.M	Developer	Integrate payment gateway and manage transactions

Dinod P.A.S.	Developer	Build delivery tracking and customer notification system
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5.3 Development Phases and Timeline

Sprint Backlog

User Story 1

Task ID	Task	Est. Hours
T-01.1	Design search interface with category filters	5
T-01.2	Implement backend search API for products	7
T-01.3	Enable search by product name, category, and brand	4
T-01.4	Add sorting by price, rating, and availability	3
T-01.5	Test search functionality with sample products	3

User Story 2

Task ID	Task	Est. Hours
T-02.1	Design shopping cart UI	4
T-02.2	Implement backend API for cart management	6
T-02.3	Allow quantity updates and item removal	3
T-02.4	Calculate subtotal, tax, and total price	3

User Story 3

Task ID	Task	Est. Hours
T-03.1	Integrate payment gateway API (e.g., PayPal/Stripe)	8
T-03.2	Implement encryption for payment data	6

T-03.3	Create payment confirmation page	3
T-03.4	Handle payment failure and retry flow	4
T-03.5	Test payment transactions	4

User story 4

Task ID	Task	Est. Hours
T-04.1	Design delivery tracking UI for customers	5
T-04.2	Implement backend to update delivery stages	6
T-04.3	Enable push/email/SMS notifications for updates	5
T-04.4	Test delivery status notifications	3

6. Conclusion & Future Work

6.1 Summary of Achievements

The **QuickMart Web-Based Supermarket Ordering System** was successfully designed and developed to simplify and digitalize supermarket operations. The project achieved its main goals by creating a secure, user-friendly platform for online shopping, order management, and delivery tracking.

Key accomplishments include implementing **secure authentication, real-time inventory updates, online payments, and delivery coordination features**. The system also supports **promotional campaigns, automated notifications, and report generation**, ensuring efficiency for both customers and staff.

Overall, QuickMart effectively meets its objectives of enhancing customer convenience, improving operational efficiency, and supporting the digital transformation of supermarket services.

6.2 Challenges Faced

- 1. UI/UX Design:** Designing an interface that was both attractive and easy to navigate for all user types.
- 2. Team Coordination:** Managing tasks and timelines effectively during Agile sprints
- 3. Database Management:** Handling relationships between multiple modules such as orders, products, and users without data conflicts.

6.3 Suggestions for Improvement or Extension

1. Technical Improvements

- Enhance security
- Optimize performance
- Use cloud hosting:

2. Feature Extensions

- Add new user roles or modules
- Enable real payment gateways
- Feedback and review system
- Notification system

3. User Experience Enhancements

- Improve interface design
- Personalized recommendations

4. Data & Analytics

- Add reports and dashboards
- Machine learning features

5. Maintenance & Integration

- Automate testing and deployment
- API integration

7. Individual Contribution, Teamwork & Lessons Learned

Member Name	Role / Responsibilities	Challenges Faced	How Challenges Were Overcome	Key Lessons Learned	Reflection (What went well / What could improve)
Jayalath J.G.H	I developed the user management portal, which handles user roles by assigning them.	Managing user data consistency across modules and ensuring admin privileges were securely handled without affecting normal users.	Implemented proper role-based authentication and access control using Spring Security	Gained a strong understanding of user role management, secure authentication, and maintaining data accuracy within admin functionalities.	The admin panel worked effectively and simplified overall management, but could be improved with more analytics features and real-time activity monitoring in future updates.
Alahakoon B.M.B.S	I developed the Inventory Management module, which included adding, updating, and tracking product stocks, monitoring low inventory levels, and ensuring real-time synchronization with customer orders.	Faced difficulties maintaining accurate stock levels when multiple users performed actions like orders or cancellations simultaneously.	Used concurrency control and transaction management to ensure stock updates remained consistent and accurate under multiple simultaneous operations.	Gained practical experience in managing database consistency and concurrency, as well as the importance of efficient data synchronization.	The inventory module worked reliably and improved order accuracy, but future enhancements could include automatic supplier notifications for low-stock items.
Dinod P.A.S	I developed the Delivery module, which handles managing deliveries, assigning drivers, tracking delivery status, and ensuring	I faced challenges with optimizing delivery assignments, tracking the real time delivery status, and managing	These issues were overcome by implementing a structured delivery assignment logic, validating the	I learned the importance of integrating delivery tracking with order and driver data for	The module integration and logic design worked well but could be improve by adding live tracking features and

	timely delivery operations.	the delivery delays.	address data, and improving the coordination with the driver and order modules.	efficiency and reliability.	automated notifications.
Sanvidu M.G.M	I developed the Payment Portal module, which included payment processing, transaction management, and refund handling.	I faced challenges with handling failed payments and refunds and ensuring secure multi-method transactions.	Implementing error handling and validation for failed payments and using Spring Boot's transaction management for secure operations.	I learned the importance of secure transaction handling, proper database integration, and frontend-backend coordination.	The payment portal functioned well after debugging and integrated smoothly, though future improvements could add real payment gateway support and better error handling.
Fernando S.A.W	I developed the Customer module, which included customer registration, login, profile management, and order history viewing features.	Managing customer registration, login, and profile updates while maintaining data security and a smooth user experience.	Used input validation, password encryption, and session management to ensure secure and reliable customer interactions.	Learned how to handle user authentication safely and design a customer-friendly interface that balances usability and security.	The customer module worked smoothly with secure login and registration features, though future improvements could focus on enhancing profile customization and adding account recovery options.
Thahani N	I developed the Marketing Management module, which	Faced challenges in scheduling time-limited	Implemented automated scheduling and validation	Learned the importance of synchronizing backend logic	The promotional features functioned

	included managing promotions, discounts, and advertisement updates to enhance customer engagement and sales visibility.	promotions and ensuring that discounts were correctly reflected across all relevant pages.	logic in the backend to control promotion periods and used dynamic rendering to update changes instantly.	with UI updates and handling time-based data effectively.	smoothly and increased system interactivity, though future improvements could include integrating analytics to measure promotion effectiveness.
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8. References

MySQL Documentation.
<https://dev.mysql.com/doc/>

9. Appendix

- Additional diagrams (ER, sequence, activity).
- Sprint logs or planning documents.